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BEFORE YOU BEGIN THE SOFTWARE LOAD - CAUTIONS AND WARNINGS

Important: If your system is connected to a Hospital Network, disconnect it before proceeding. Failure to disconnect the SGI computer can corrupt the software load process.

Important: Remove power from the DASM before beginning the load; remember to restore power after the load has finished. The OnLine Center Support has reported intermittent system lockups on Indigo computers during software loads with the DASM connected and from time to time the Octane computer will not load with the DASM running.

Important: During the Install GUI phase of the software load, be sure the **<Num Lock>** key on the keyboard is turned off. If the **<Num Lock>** is on, the symptom seen is that nothing will happen when trying to type in any of the edit fields in the Install GUI.

Important: When reconfiguring the system using the Install/Configuration GUI, remember to push the **<Configure>** button on the verify tab before going to the Save info tab and doing a SaveInfo. If you have not done the final configuration by pushing the **<Configure>** button in the Verification tab, the information saved during Save Info will not reflect the latest changes made in the Install GUI.

Important: If at any time during the OS load a problem occurs that causes you to stop the install, correct the problem then reset the computer or power cycle the computer and begin the OS install again from the start. This is necessary to insure a complete load. If this does not fix your problem it may be necessary to do a low level format on the system drives. Refer to Appendix C for this process.

Important: If at the end of OS load the system will not boot and stops with the error, "/var/root partition corruption". Do the following:

1. Open a shell window by typing **sh <Enter>**.
2. Type at the prompt: **dd if=/dev/zero of=/dev/rdsk/dks0d1s0 count=1000 <Enter>**.
This writes zeros across the disk and will run for less than one minute.
3. Start the OS load from the beginning.

This error is a very rare event.

1- LOADING HOST COMPUTER - OVERVIEW & PREPARATION

Note

This procedure covers Host software install for Release 8.3 M3 and 8.4 M4.

IMPORTANT!

If this is a new LX install or system upgrade to a Signa LX Cardiac (CV) or Neuro (NV) system, the system should be pre-loaded with Release 8.4 software when the M4 version (Product Release) is available. For this case, refer to procedure "Signa Release 8.3 and 8.4 Host Reconfiguration" (OW2SCAE.DOC). If the system was pre-loaded with Release 8.3 software, it must be loaded with Release 8.4 software using this procedure.

1-1 Required Media

- IRIX 6.5 Operating System CDROM Set, Disk 1 of 2 and Disk 2 of 2
- IRIX 6.5.4 Boot/Update CDROM
- MrpApps - MR Applications CDROM
- MrpResSrv - MR Restricted Service CDROM (*GE Service use only*)
or MrpAdvSrv - MR Advanced Service CDROM (*Licensed In-House Customers*)
- saveINFO - MaxOptix MOD Media (Either blank or previously used)

DO NOT PROCEED UNLESS YOU HAVE ALL THIS MEDIA AVAILABLE.

1-2 Archive Images

This installation instruction will erase any current images left in the database. **Archive all images BEFORE continuing with this procedure.**

1-3 Required Information Prior To Loading Software

First collect vital site information listed below to insure the install will not be stopped for lack of site specific information (and in case RestoreInfo fails). To collect this information:

1. On the Service Desktop, select **[Install]**; type **operator <Enter>** for the password.
2. Look in each tab for current site information. (*For new installs, refer to Appendix A for descriptions and instructions on how to obtain certain Network and System Configure site information.*) At a minimum, the "vital" entries (those with a checkmark (✓)) should be recorded in Table 1-1. However, it is recommended that you record all site configuration information listed in Table 1-1.

TABLE 1-1
SITE CONFIGURATION INFORMATION

TAB	ENTRY (✓- Vital Info)	VALUE
Network Info	✓ Host IP Address:	
	✓ Netmask:	(Default is 255.255.252.0)
	✓ Hostname:	
	✓ Domain Name:	
	YP(NIS) Service:	☐ On ☐ Off
	✓ IWS PC IP Address:	
	✓ FE Laptop IP Address:	
	✓ TPS Subnet (IP):	
	✓ DICOM Network Timeout:	☐ Slow ☐ Medium ☐ Fast
System Configure	✓ Hospital Name:	
	✓ Suite ID:	
	✓ Host ID:	
	✓ Unique ID:	
	✓ Service ID:	
	Date Format:	
	Doctor Title:	☐ Radiologist ☐ Diagnostician
	Weight Unit:	☐ Pounds ☐ Kilograms
	Monitor Type:	☐ Gray ☐ Color
	Palette:	☐ Gray ☐ Color
	Keyboard:	☐ Amer. ☐ Fr. ☐ Ger. ☐ Ital. ☐ Port. ☐ Span. ☐ Swed.
	Language:	☐ Amer. ☐ Fr. ☐ Ger. ☐ Ital. ☐ Port. ☐ Span. ☐ Swed.
Hardware Configure	Line Frequency:	☐ 60Hz ☐ 50Hz
	✓ Field Strength:	☐ 1.0T ☐ 1.5T
	✓ Package Name:	☐ EchoSpeed SR150 ☐ EchoSpeed SR120 ☐ HiSpeed SR77 ☐ Base (Horizon) SR20
	✓ Gradient Type:	☐ 8645 Single or Double Bay ☐ 8280 SGD Base System ☐ 8651 SGD HighSlew ☐ 8915
	✓ RF Amp Type:	☐ Tube Type RF PEN II ☐ 1.0T Teli Solid State RF ☐ 1.5T Analogic Solid State RF
	ISO Vector Z:	☐ 8981 ☐ 12520 ☐ 15250
	Magnet Serial Number:	
	Table Limit:	
	✓ CERD (KHz):	☐ CERD 1/4 Ch 62.5 62.5 62.5 62.5 (See Note 1 below for explanation) ☐ CERD1/4 Ch 125 125 125 125

TAB	ENTRY (✓- Vital Info)	VALUE
	<i>of values)</i>	___ CERD 4 Ch 125 62.5 62.5 62.5 ___ CERD 4 Ch 125 125 62.5 62.5 ___ CERD 4 Ch 125 125 125 62.5 ___ UCERD 1/4 Ch 125 125 125 125
	✓ Resonance Module:	___ BRM ___ CRM
Edit Patient Data	Edit Patient Data Time Limit:	
	Max Images Installed Per Update:	
	Log File Header:	Log File Header (Default)
	Date Format:	___ Month/Day/Year ___ Day/Month/Year
Set Exam Number	✓ Exam Number (1-49999):	
	Diag Exam Number (50000-65535):	
Set Disk Limit	Set UDDM (60-100%)	
DASM	DASM Interpolation	___ Linear ___ Cubic
	✓ Laser Camera Type:	___ 3M952 ___ 3M XL or HQ ___ 3M DryView ___ AFGA MIN ___ Kodak ___ Dupont ___ Fuji ___ Konica
	✓ Install DASM?:	[] No [] Yes
VCR Setup	VCR Format:	___ NTSC ___ PAL
HIS/RIS	Option Names:	[] Current Patient Location [] Special Needs [] Pregnancy Status [] Referring Physician [] Request Contrast Agent [] Contrast Allergy [] Medical Alert [] Additional Patient History
	✓ HIS/RIS (Server) AE Title:	
	✓ HIS/RIS (Server) IP Address: If Option Is Installed	
	✓ HIS/RIS (Server AE) Port #:	
	✓ MR Server AE Title:	
	✓ HIS/RIS Patient Name Delimiter:	[] No [] Yes
Set Time/Date	Time Zone:	
--	Modem Type: <i>(Not in Install tab; look at modem)</i>	

Note 1: In the **Hardware Configure** tab, the CERD choices are as follows:

- **CERD 1/4 Ch 62.5 62.5 62.5 62.5** – for single 62.5 KHz, or four 62.5 KHz Receivers
- **CERD 1/4 Ch 125 125 125 125** – for single 125 KHz, or four 125 KHz Receivers
- **CERD 4 Ch 125 62.5 62.5 62.5** – for 125 KHz Receiver 1, and 62.5 KHz for other Receivers
- **CERD 4 Ch 125 125 62.5 62.5** – for 125 KHz Receivers 1, 2, and 62.5 KHz for Receivers 3, 4
- **CERD 4 Ch 125 125 125 62.5** – for 125 KHz Receivers 1, 2, 3 and 62.5 KHz Receiver 4

- **UCERD 1/4 Ch 125 125 125 125** – for single UCERD, or 4 channel UCERD.
If your system configuration does not fit any of the selections above (e.g. 62.5 KHz for Receiver 1, and 125 KHz for any other Receiver), move the higher bandwidth Receiver(s) to the lower slots beginning with Receiver 1; this will put the higher BW on the most used Receivers. Then select the entry that matches your final configuration.
3. Check (✓) the Options installed by selecting **Option Keys** tab from the Install GUI, and recording them in Table 1-2. After loading software and doing Restore Info, verify that all the options are still available to the customers. If any keys are missing, reactivate them using the option key floppy, or MOD.

TABLE 1-2
SYSTEM OPTIONS DATA TABLE

SITE OPTIONS		SITE OPTIONS
Echo Planar Imaging		Tagging
Fast Gradient Echo		Sgdperf
Cine		DW-EPIEZ
Fast Spin Echo & FLAIR		iDrive Pro
Time of Flight		iDrive
Phase Contrast Vascular Imaging		SmartPrep 2000 Upgrade
Research		Probe 2000 Upgrade
Proactive Service		Multi-Nuclear Spectroscopy
Research PSD		Functool
Spectroscopy/PROBE		Voxtool
Videoconferencing		Interavascular Imaging
SGD_EchoSpeed		Clariview
DW EPI	x	Multislice Multiangle (Note 1)
FLAIR EPI		Fgret (Note 2)
SPECIAL		fgre Real Time (Note 2)
Smart Prep		Spiral High Resolution (Note 2)
SSFSE		Spiral Real-Time (Notes 2 & 3)
Three Plane Localizer		iDrive Pro Plus (Notes 2 & 3)
HIS RIS		Spectro Analysis GE
E3D TOF		
FSE_XL		
Bloodsupp		
Fastcine		

- Note 1:** All M4 customers receive the Multislice Multiangle option at no cost. No option key is required; the option is automatically installed each time you login as “Signa” with Release 8.4 M4 software.
- Note 2:** This option is located on the Signa CV 8.4 Cardiac Option Keys MOD.
- Note 3:** This software/option is not available for customers until 8.4 M4 (Product Release)

1-4 SaveInfo and Saving System Information Entered In The Install GUI

1-4-1 SaveInfo

A SaveInfo should be done before starting the install to save the latest system configuration files, calibration files, and protocols.

1. Insert blank MOD or flip side of existing Info MOD.
2. From the Service Desktop select **[Install]**. In the Install Tool window, type **operator** **<Enter>** at the password prompt.
3. In the Install GUI, select the **Save/Restore** tab.
4. Select **[Save Information]**. This saves all system calibration and configuration files, along with customer protocols.
5. When SaveInfo is complete, you can try the following command to give you an indication that the SaveInfo was successful. Open a C Shell on the Service Desktop. Login as *root* by typing: **su root**, then enter password **operator**.

Type the following:

tar vtf /MOD_DRIVE <Enter> (there is a space between vtf and / character)

Hundreds of files that were backed up on the MOD should scroll by at this point. This will be a good indicator that the SaveINFO was successful. Note: to view the files a page at a time you can type: **tar vtf /MOD_DRIVE|more <Enter>** (the vertical character “pipe” followed by the word “more” will display a page of output at a time; use the space bar to continue paging).

6. When SaveInfo is complete, remove the MOD media and write **8.4 Info MOD**, the system name, and the date on it.

1-4-2 Save GI Configurations

The following will save the Guided Install configuration to the Info MOD created above.

1. Re-insert the Info MOD created above (same side).
2. At top left corner of the Install GUI, select **<File>** then **<Save GI Configuration to MOD>**.

Note

The “Save GI Configuration to MOD” does not create a complete Info MOD. It only creates a copy of the information already entered in the GUI tabs for later use to update the tabs during a software reload. The **<Save GI Configuration to MOD>** and SaveInfo can be saved on the same Info MOD. A separate partition is built for each.

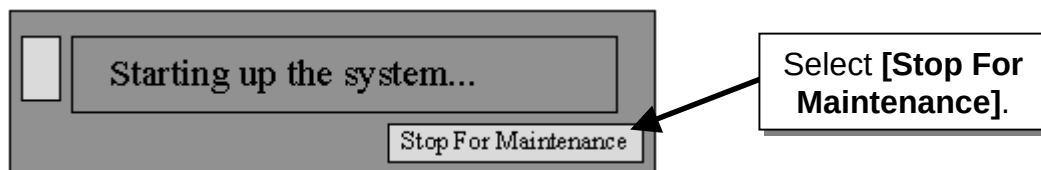
2- HOST OPERATING SYSTEM SOFTWARE LOAD

Initial Conditions:

- All images archived (Section 1-2).
- System information recorded (Section 1-3); SaveInfo performed (Section 1-4).
- System disconnected from Hospital Network.
- DASM power removed (per "Before You Begin..." important note for DASM).

Procedure:

1. If the system is not up and running, switch on power at this time and continue with step 2. If the system is already running, go to the Service Desktop and select **[Shutdown]**. When prompted, click on **[Restart]**.
2. Stop the reboot process; click on **[Stop For Maintenance]** when it appears.



3. Click on **[Enter Command Monitor]** when it becomes available.
4. Insert the **IRIX 6.5** CD labeled "**Boot/Update**" into the CDROM Drive on the workstation (right side). On the command line type the following (*user responses are bolded text*):

Depending on the host computer type your system is using, boot the CD by typing one of the following:

For SGI Octane computer:

```
>> boot -f dksc(1,3,8)sash64 dksc(1,3,7)stand/fx.64 --x <Enter>
```

For SGI Indigo computer:

```
>> boot -f dksc(0,3,8)sashARCS dksc(0,3,7)stand/fx.ARCS --x <Enter>
```

Note

The rest of the install is for both SGI computer types.

```
fx: "device-name" = (dksc) <Enter>
```

```
.
```

```
fx: ctrlr# = (0) <Enter>
```

```
.
```

```
fx: drive# = (1) <Enter>
```

```
.
```

```
fx> r/ro <Enter>
```

```
.
```

```
fx/Repartition/RootDrive: Type of Data Partition = (xfs) <Enter>
```

```
.
```

```
fx> continue? yes <Enter>
```



```
.  
fx> l/sy <Enter> (This is a lower case l , l, lower case s, and lower case y.)  
.  
fx> exit <Enter>
```

5. At this point the system will reboot to the PROM startup screen. Select **[Install System Software]**. A pop up window opens and asks you where to install from. Choose **[Local CD-ROM]**, select **[Install]**, then select **[Continue]**.

During this boot, the system may tell you it was unable to mount the “/root” file system. The system will stop with the statement "press Enter to invoke C shell csh:". If you see this message, type the following:

```
...press Enter to invoke C shell csh: <Enter>  
# mkfs /dev/dsk/dks0d1s0 (This is /dev/dsk/dks zero d one s zero.)  
# exit <Enter>
```

(This problem occurs with newly formatted disk drives or new, never used disk drives, but has also been seen with old drives during software reinstall.)

6. The next system message should be “Trying again to mount /dev/dsk/dks0d1s0 on /root.” If you get this message, continue with step 7. Otherwise...

If you get a message similar to:

```
/dev/dsk/dks0d1s0: Invalid argument No valid file system  
found on: /dev/dsk/dks0d1s0 This is your system disk:  
without it we have nothing on which to install software.
```

Do the following:

```
Make new file system on /dev/dsk/dks0d1s0 [yes/no/sh/help]: yes<Enter>  
About to remake (mkfs) file system on: /dev/dsk/dks0d1s0.  
This will destroy all data on disk partition: /dev/dsk/dks0d1s0.  
Are you sure? [y/n] (n): y<Enter>
```

```
Block size of filesystem 512 or 4096 bytes? 4096<Enter>  
Doing: mkfs -b size=4096 /dev/dsk/dks0d1s0  
meta-data=/dev/rdsk/dks0d1s0 isize=256 agcount=8, agsize=128612 blks  
data = bsize=4096 blocks=1028895, imaxpct=25  
log =internal log bsize=4096 blocks=1000  
realtime =none bsize=65536 blocks=0, rtextents=0  
Trying again to mount /dev/dsk/dks0d1s0 on /root.
```

7. Continue as follows:

```
inst> sh <Enter>  
.  
# mount /CDROM <Enter>  
.  
# /CDROM/mr_prep <Enter>  
.
```

Note

During the software load, there will be much screen output and many pauses of 10 to 30 seconds, no entry is required at these pauses. Approximate wait times are listed for any lengthy pauses, wait for the prompts as detailed in the following steps.

8. The system will run for about 6 minutes (Octane) to 15 minutes (Indigo) and then will prompt with:

"Please insert the First OS Disk (Pa) and then press <enter>".

Put the OS CDROM Disk 1 of 2 into the CDROM Drive. Wait for 20 seconds, or until the amber light on the CD Drive stops blinking, then push the **<Enter>** key.

9. After an additional 6 minutes (Octane) to 15 minutes (Indigo), the system prompts with:

"Please Insert the Second OS Disk (Pb), then press <enter>"

Put the OS CDROM Disk 2 of 2 into the CDROM Drive. Wait for 20 seconds, or until the amber light on the CD Drive stops blinking, then push the **<Enter>** key.

Loading the rest of the Operating System off the second disk will take about 25-30 minutes and will not require any user intervention.

10. When finished, the system will stop at the "#" prompt. Type the following to continue.

.
exit <Enter> (to exit back to the original inst> prompt.)

.
.
Inst> **exit <Enter>**

.
Automatically reconfiguring the operating system..

.
.
.....Ready to restart.... **y <Enter>** (Restarts the system)

.
The system will finish some processing and reboot. The OS load portion of the Software Install is complete.

Important!

Do not log into EZ Setup. It is no longer used as part of Install.

Important: If at any time during the OS load a problem occurs that causes you to stop the install, correct the problem then reset or power cycle the computer and begin the OS install again from the start. This is necessary to insure a complete load. If this does not fix your problem it may be necessary to do a low level format on the system drives. Refer to Appendix B for this process.

11. Continue to Section 3 of this document for the MR_Apps software load.

3- MRP_APPS AND SERVICE SOFTWARE LOAD

3-1 Load Applications Software

1. When the system is finished rebooting, log in as **root**. There is no password at this time. You will first see a startup box that will asks for the MrpApps CDROM. See Illustration 3-1.



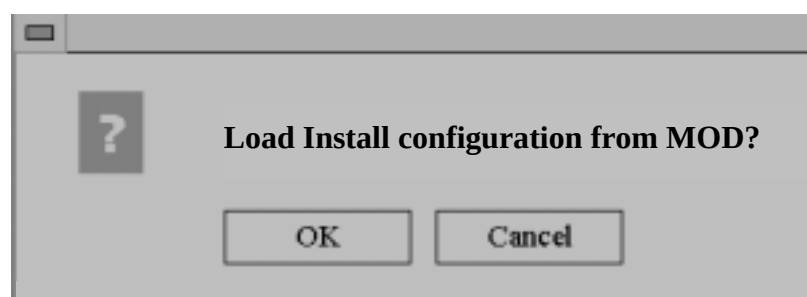
GUIDED INSTALL LOAD SCREEN
ILLUSTRATION 3-1

2. Remove any CD from the CDROM drive and insert the MrpApps CDROM into the CDROM drive. Wait 20 seconds or until the amber light on the CDROM drive stops blinking, then select **[Load]**. The Guided Install will load and start.

Note

Help documentation is built into the Guided Install GUI. Select **[Help]** on the tab you need information for in the GUI. When the GUI first starts, select **[What's New]** for information on using the GUI.

3. On initial startup, a popup box will appear over the Guided Install interface describing the tool and its use. After reading the information select **[OK]** to continue.
4. Next, a popup box (see Illustration 3-2) will appear over the Guided Install interface. Insert the **8.4 Info MOD** created in Section 1-4-2 to restore the previously "Save GI Configuration to MOD" information. If this MOD is not available, select **[Cancel]** to continue manually with the install. *(The normal SaveInfo files are not restored at this point in the software load. They will be restored in step 10.)*



INSTALL CONFIGURATION POPUP SCREEN
ILLUSTRATION 3-2

5. The Guided Install home screen is displayed. See Illustration 3-3.



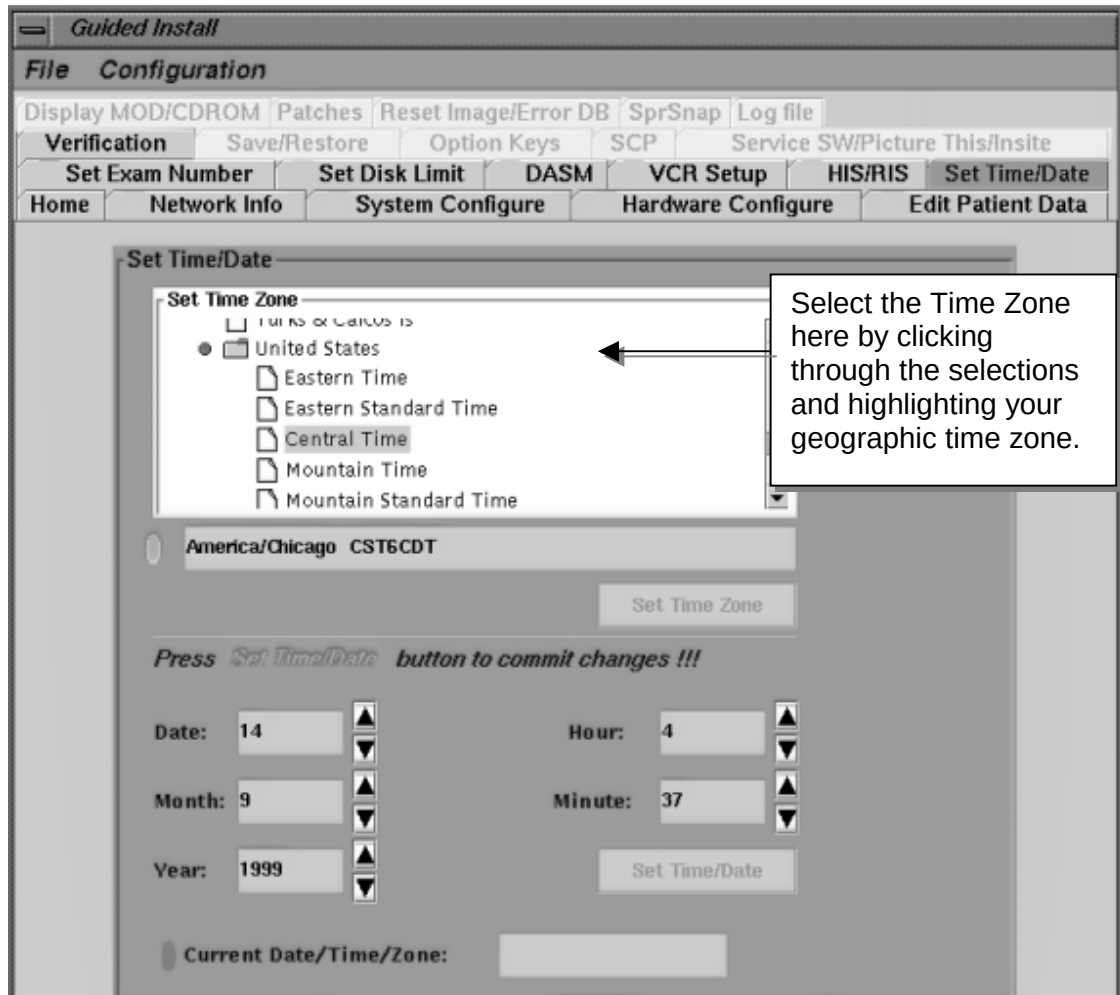
GUIDED INSTALL - HOME SCREEN
ILLUSTRATION 3-3

6. Press the **<Next Tab>** button to move through all active tabs from "**Network Info**" to "**Set Time/Date**". If your **Save GI Configuration MOD** was used, verify the information in the tabs. If not fill out all empty entries or make any updates as appropriate. Use the information you recorded in Table 1-1 to verify/make entries.

- When you reach the "Set Time/Date" tab, go to the "Set Time Zone" box and use the selections to set the time zone your system is in. See Illustration 3-4. (Correct entries need to be made in the Network and time zone tabs.)

IMPORTANT!

Correct entries need to be made in the Network and Time Zone tabs. The entries in the other tabs need to be filled out, but the values from the Info MOD will replace your entries when the RestoreINFO is done following the applications software load.

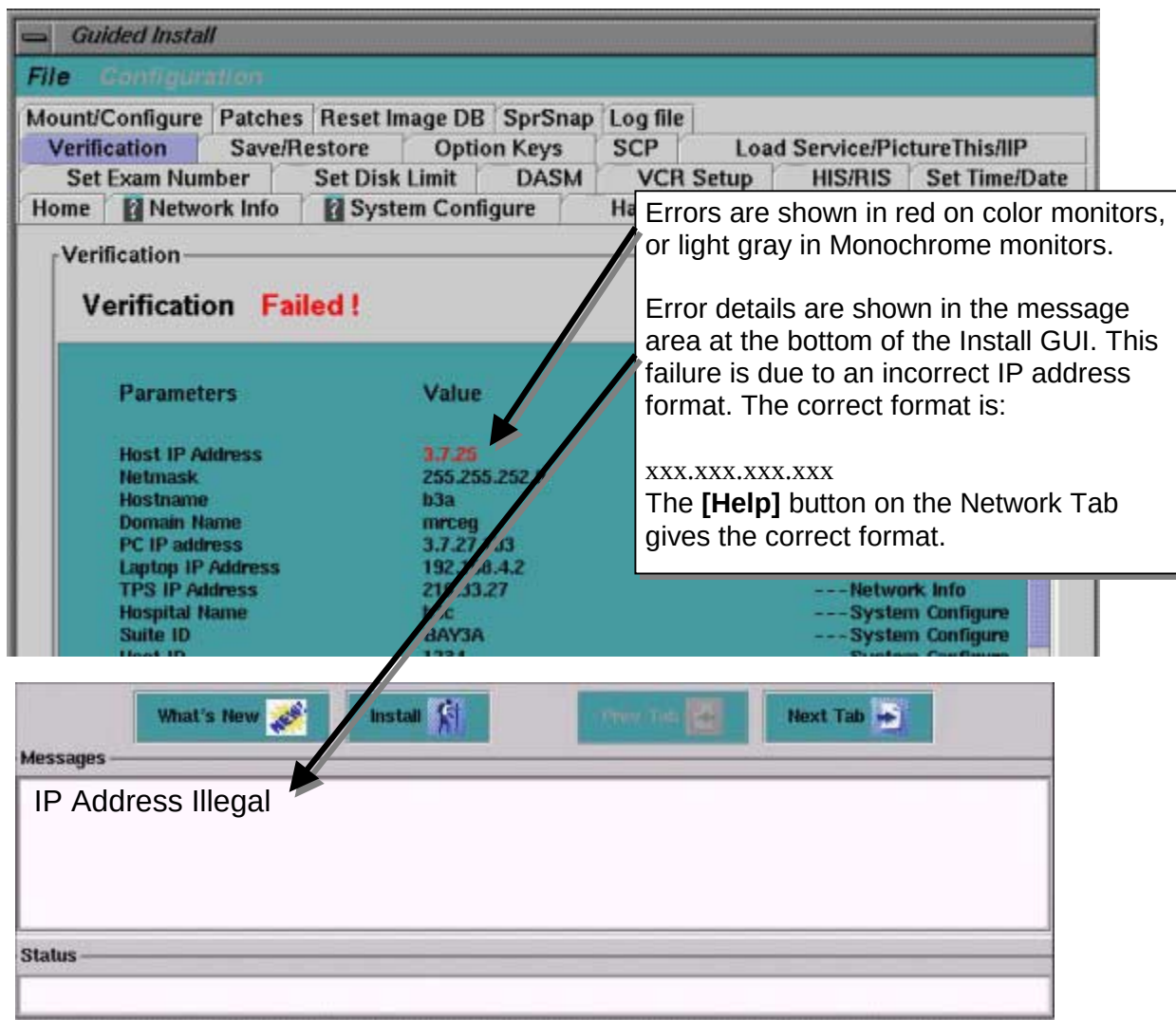


GUIDED INSTALL - SET TIME/DATE SCREEN
ILLUSTRATION 3-4

- Proceed to the **Verification** tab. Once in the Verification tab, look at the pass/fail status of the entries. If any entry has a failed status, the **[Install Software]** button will not be active. Failed entries will be due to missing entries, or entries not in the correct format. Go to the tab with the error and correct it, then select **[Install]** to open the **Verification** tab again. If all entries pass verification at this point, select **[Install Software]** to begin the MrpApps load. The MR Apps load will take about 20 minutes for Octane and 40 minutes for Indigo.

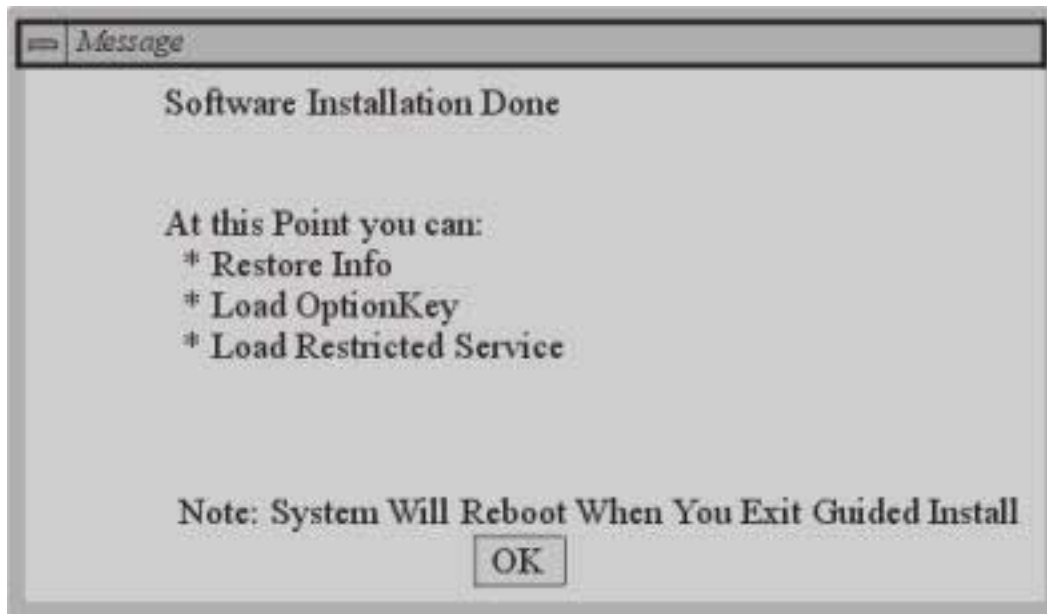
Note

It is not necessary that any one tab be filled out in any particular order. Just all required configuration tabs must be filled in or the install will not continue. The Verification Tab will list any formatting errors as well as current settings. See Illustration 3-5 for an example of a failure in the Verification Tab in networking (in the example, the Host IP Address is an incorrect format).



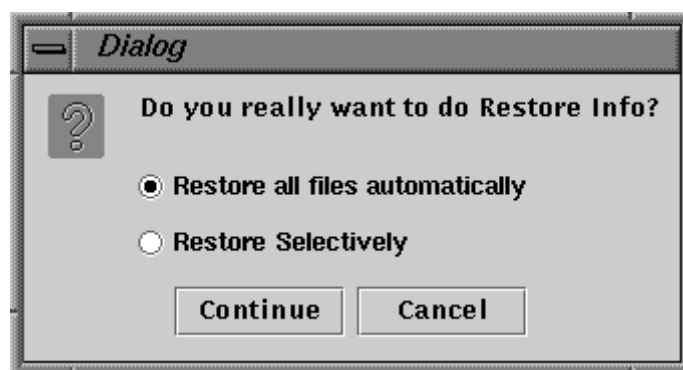
GUIDED INSTALL - VERIFICATION SCREEN
ILLUSTRATION 3-5

9. Once the MR Apps load is done, the popup in Illustration 3-6 is displayed. Select **[OK]** to close the popup.



GUIDED INSTALL – SW DONE SCREEN
ILLUSTRATION 3-6

10. If the site has *Voxtool*, *Functool*, *Interactive Vascular Imaging* or *Clairview* options, go into the Guided Install GUI, select **Option Keys** tab and reload these options at this time from the original MOD's as they are not saved with SaveInfo. Loading these options now will allow customer generated protocols for any of these options to load later during RestorInfo.
11. At this point the GUI will change from the Install phase to the Configure phase. Insert the Info MOD created in Section 1-4-1, select the **Save/Restore** tab and then select **[Restore Information]**.
12. A popup will ask if you want to restore all files or selectively restore files. See Illustration 3-7. Select **Restore all files automatically**, then **[Continue]**.



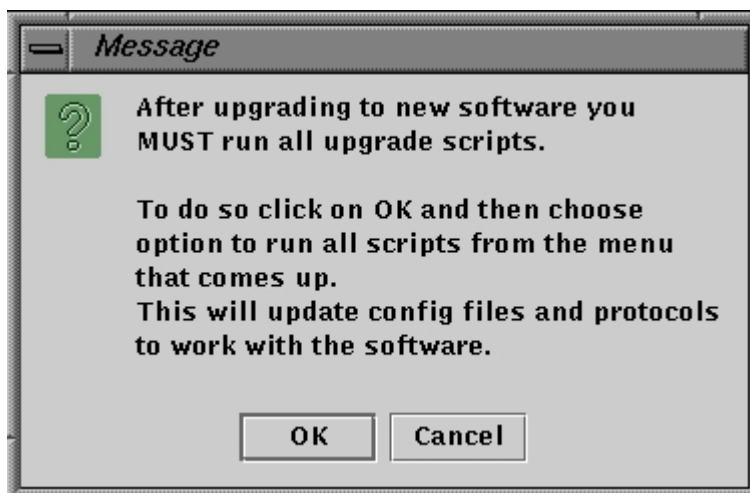
RESTORE OPTION POPUP
ILLUSTRATION 3-7

13. A “RestoreINFO in Progress” window will scroll filenames as they are being restored and then a popup will ask if you want to specifically restore the MRconfig file. See Illustration 3-8. Select **[Yes]**.



MRCONFIG FILE RESTORE POPUP
ILLUSTRATION 3-8

14. When the RestoreInfo is finished a popup will open telling the installer that upgrade scripts must be run. See Illustration 3-9.



GUIDED INSTALL – UPGRADE SCRIPTS MESSAGE
ILLUSTRATION 3-9

15. Select **[OK]** to continue. The upgrade script menu will open (see Illustration 3-10).

This script upgrades the system config files for LFC software. After Restore Info is completed, the following UP scripts must be run before scanning.

- | | |
|--------------------|--|
| 1. ConfigUp | (Update MRconfig.cfg/GradientConfig.cfg files) |
| 2. CoilUp | (Update CoilConfig.cfg file) |
| 3. ProtocolUp | (Upgrades 82.5/83 Protocols) |
| 4. ProtFormat | (Reformats site protocols names) |
| 5. Restore_License | (Must be run individually) |
| 6. Run all scripts | (Except Restore_License) |
| 7. Exit | (Return to Guided Install.) |

GUIDED INSTALL – UPGRADE SCRIPTS MENU
ILLUSTRATION 3-10

16. You must always run these updates in the following order:

- a. Type: **6 <Enter>** *This will take about 1 minute to run.*
- b. Type: **7 <Enter>** *This will close the menu and return to the GUI.*

Note

Refer to Appendix D3 and D4 for parameters updated by the upgrade scripts

17. Once back on the Guided Install screen, use the cursor to select **<File>** and then **<Quit>** at the top left of the GUI. When the GUI shuts down, the system will reboot and all changes restored or made will be set.
18. After the system has finished rebooting, once again log in as **root**. Password is **operator**.
19. You will be asked, once entering the root desktop, if you want to start the Install GUI. Select **[Yes]**. At this point:

If your system has the VCR Interface option (there will be a VCR connected to the computer), proceed to Section 3-2, Video Driver Load for Systems with VCR I/F Option Only, to load the necessary video driver.

If your system does not have the VCR Interface option (no VCR connected to the computer), proceed to Section 3-3, Service Software Load.

3-2 Video Driver Load for Systems with VCR I/F Option Only

If your system does **not** have the VCR I/F Option proceed to Section 3-3, Service Software Load. Systems that have the VCR Interface Option must follow the steps in this section to load the video driver because it is not loaded with the operating software load. Follow these steps to install the video driver software:

1. Insert the IRIX 6.5 Boot/Update CDROM into the CDROM drive.
2. In the winterm window type: **mediad <Enter>** (to mount the CDROM drive).

3. Type: **cd /CDROM/opt <Enter>**

4. Type: **./vidsetup <Enter>**

This starts the install script. Screen output will scroll by in the window, and it will be apparent that file installs/removals are taking place. The last output displayed is:

```
Reboot to start using the reconfigured kernel.
*****
**                                                    **
**                                                    **
**      Remember to reboot the system before        **
**      attempting to use the video interface        **
**                                                    **
*****
{sdc@name}
```

5. Type the following to verify the video driver was loaded:
showprods -D1 <Enter>

examine the products that are loaded and ensure the following is listed:

```
evo <date installed> OCTANE Personal Video Execution Environ, 6.5.4m
```

6. Press the spacebar until you get a prompt back.

7. Type: **mediad -k <Enter>** (to release the CDROM device).

8. Remove the Boot/Update CDROM.

Note

Do not reboot the system at this point to activate the video driver. The system will be rebooted after loading the service software in the next section.

3-3 Service Software Load

This section applies only to GE Service and in-house customers with a valid Advanced Service Package Limited License.

3-3-1 Loading Service Software

1. Go to the install GUI window, and select the **ServiceSW/Picture This/InSite** tab.
2. **In-House Customers:** Proceed to Section 3-3-2, Load Advanced Service Software.
GE Service: Proceed to Section 3-3-3, Load Restricted Service Software.

3-3-2 Load Advanced Service Software

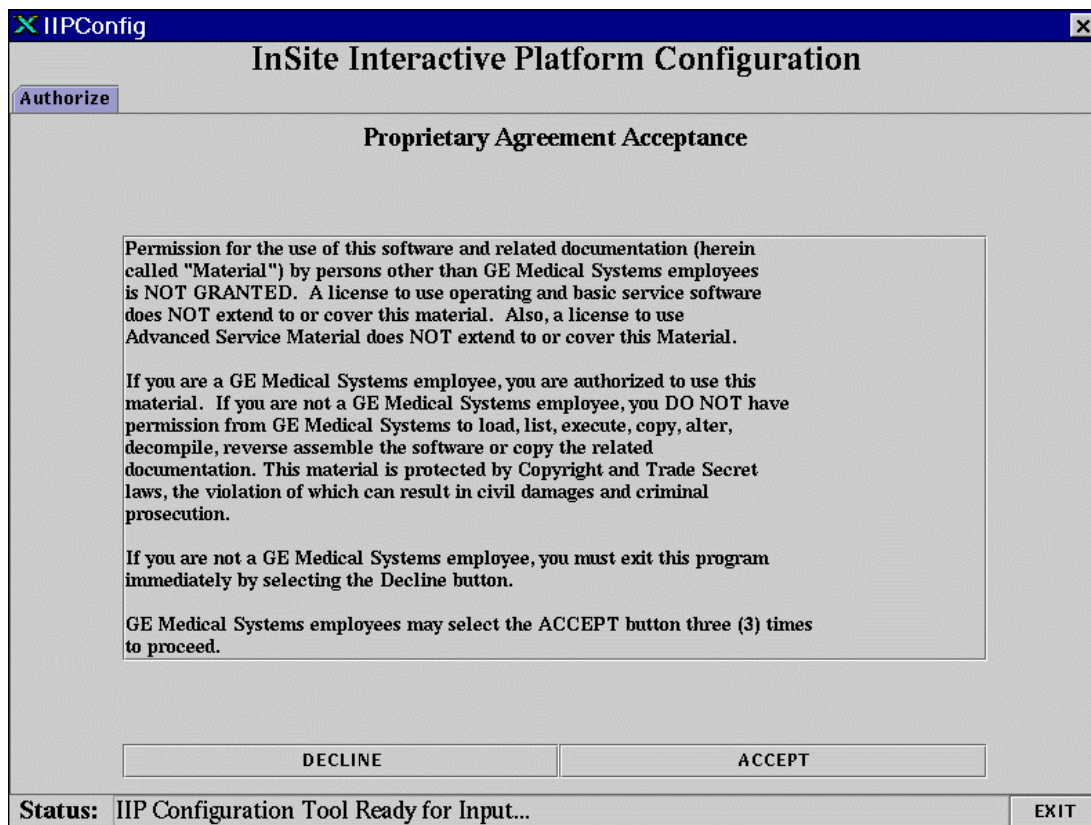
Advanced Service Software is only available to sites with a valid Advanced Service Package Limited License. To load Advanced Service Software on the system:

1. Place the Release 8.4 Advanced Service CD-ROM in the host CD-ROM drive.
2. Select **[Load Advanced Service]**.
3. Read the proprietary message window, and respond accordingly.
4. A message "Advance software load in progress..." will display. The load will take about 5 minutes.
5. Wait for the message "Advanced Service load done" in the Install GUI status bar (at bottom of window).
6. Proceed to Section 3-4, Option Keys and Verification of Restored Information.

3-3-3 Load Restricted Service (for sites with GE Service only)

Restricted Service Software is only available for GE use. To load Restricted Service Software on the system:

1. Place the Release 8.4 Restricted Service CD-ROM in the host CD-ROM drive.
2. Select **[Load Restricted Service]**. Read the proprietary message and respond accordingly.
3. A message "Restricted software load in progress..." will display. The load takes about 14 minutes; be patient. Illustration 3-11 displays near the end of the Restricted software load.



PROPRIETARY AUTHORIZATION TAB

ILLUSTRATION 3-11

4. GEMS Field Engineers loading GEMS Restricted Service Software should also configure the InSite Interactive Platform at this time (see Illustration 3-11). Select **[Accept]** three times, then follow the instructions in the popup windows to setup the **[Information]**, **[ProDiags]**, **[Health Page]**, **[Device Connect]**, and **[InSite Checkout]** tabs. On the **InSite Check** tab, you can select **[AUTO CHECKOUT]**. During InSite Checkout you may be asked for System Model type. For release 8.4 the model type is LX-IIP2.

Note

For information on the InSite Interactive setup, refer to the procedure "Insite/IIP Installation Signa 8.3 or Above" on the Signa 8.x Service Methods CD-ROM (2160623-1), under "System" procedures.

Note

Pro Diags are defaulted to call out to the ASC daily. If the attempt to call out fails, the software reattempts to call out hourly until successful. If the modem or network connection is down, **these hourly attempts to call out can slow down the scanner**. It is suggested that during InSite IIP setup, ProDiags call out to ASC be configured to dial out weekly and reattempt daily

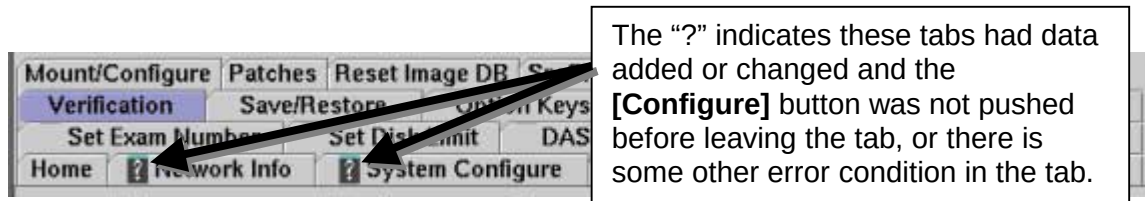
3-4 Option Keys and Verification of Restored Information

1. Refer to the list of options you recorded in Table 1-2 and verify all options are listed as "permanent" in the Option Key window. If any options that were recorded in Table 1-2 are listed as "uninstalled", reload them from the original MOD's.

2. Move through and check the accuracy of the entries that were restored in each tab of the Install GUI. If corrections/changes are required, enter the changes, then select the **[Configure]** button on the bottom of the tab to apply the change. If no changes are required, proceed to next step.



If **[Configure]** is not pushed to commit changes to a tab, or an error is present in a tab, a question mark will be seen on the tab, as shown in Illustration 3-12.



CONFIGURATION PROBLEM EXAMPLE
ILLUSTRATION 3-12

3. Perform a SaveInfo to save the latest changes by inserting the site Info MOD or blank MOD into the drive, going to the **Save/Restore** Tab, then selecting **[Save Information]**.
4. At the top left corner of the Install GUI, select **<File>** then **<Save GI Configuration to MOD>**. This will save information that has been entered in all the edit fields in the GUI.

Note

The “Save GI Configuration to MOD” process does not create a complete Info MOD. It only creates a copy of the information already entered in the GUI tabs for later use to update the tabs during a new install. The <Save GI Configuration to MOD> and SaveInfo can be saved on the same Info MOD. A separate partition is built for each.

5. Exit the install GUI by selecting **<File>** then **<Quit>** from the top of the interface. If any error conditions still exist, you will be warned that no changes will be made. If no error conditions are found, the GUI will save user parameters.

3-5 Hardware Reconnect, RTIP Re-Enable, and System Reboot

1. Reconnect the following hardware at this time if it was disconnected at the beginning of the install:
 - Network Interface
 - DASM power
2. **Sites without Neuro RTIP Option on AW:** Skip this step; continue with step 4.
Sites with Neuro RTIP Option on AW: The “update_signa” **must** be re-run anytime Signa software is reloaded. RTIP will not be functional following a Signa reload until “update_signa” is run from the AW.

After logging in on the AW as sdc:

- Iconify the AW/VP's browser (click the left mouse button on the upper-right corner of the browser).
- Open a command window (using the root menu, brought up by clicking the right mouse button anywhere in the background, and selecting the "command window" option from the "service tools" menu).
- Type **su** in the command window and enter the root password (**operator**).
- Type: **cd RTIP/INSTALL <Enter>**
- Type: **./update_signa** (note the leading "." - see below) and follow the prompts to complete the update. Here is a transcript:

b5-aw# **./update_signa**

b5 is alive *(Example value 'b5' for Signa hostname shown)*

Enter Signa root password: **← Enter site's Signa root password** (default is **operator**).

Validating, please stand by... done.

Enter Signa user password (user 'sdc'): **← Enter site's Signa sdc password**
(default is **adw2.0**)

Validating, please stand by... done.

svat_hosts.cfg not found on signa host - will attempt to install it.

Installed correct svat_hosts.cfg onto the signa.

AW not entered in the signa 'hosts' file; will attempt to add it...
...done.

AW not entered in signa 'exports' file; will attempt to add it...
...done.

Exporting signa's /export/home1/sdc_image_pool/mr_scan file system...

Successfully exported signa's /export/home1/sdc_image_pool/mr_scan dir to AW
(got successful status from exportfs)

checking status returned from lais (host is authorized)...

==> lais is running properly <==

Using signa hostname 'b5'... *(Example value shown)*

Adding Signa hostname 'b5' to /etc/auto_home...

...done. The original file has been moved to /etc/auto_home.orig.

Now checking signa host 'b5'...

==> Communication with signa host 'b5' ESTABLISHED. <==

Checking tps 'b5-tps'...

===> Communication with tps ESTABLISHED. <===

Enabling the rtt init file...

Enter Signa root password: ← Enter site's Signa root password (default is operator)

Validating, please stand by... done.

...complete!

```
* * * * *
*                                     *
*                                     *
*          -----                   *
*        ---- NOTE ----             *
*          -----                   *
*                                     *
* THE SIGNA HOST COMPUTER MUST BE REBOOTED BEFORE RTIP CAN BE USED      *
*                                     *
*                                     *
* * * * *
```

3. From the Signa Toolchest window, select **Desktop; Log Out**. Then select **[Yes]**.
4. Select **reboot**, then **[Login]** to reboot Signa. For Mobile sites, first login as **mobile** to verify the mobile site information restored properly. Next, login as **Signa** (default password **adw2.0**).
5. This concludes the software load. Continue with Section 4 to update the Monitor gamma settings.

4- GAMMA SWAP & FRAME RATE, CONTRAST & BRIGHTNESS & GRAYSCALE CALIBRATION

After a complete software load has been done, proceed to the specific Host Monitor Setup and Calibration procedure for your monitor type. Failure to perform the Monitor calibration will result in Customer complaints related to Image Quality and may cause unnecessary system troubleshooting.

To make sure that LCD monitor image quality is optimized, set up the following:

- Gamma Setting (Selecting the correct monitor type in Guided install will automatically install the correct gamma table for that monitor type.)
- Swap and Frame Rate
- Contrast and Brightness
- Grayscale
- DASM (If present)

1. Choose the appropriate document:

- For color and black and white use the procedure [[Models MDW321 and MDW321+ Monitor Adjustments](#)].
- For NEC LCD monitors use the procedure [[NEC 2000, 2010, 2010X Host LCD Color Monitor Calibration](#)].
- For Eizo L660 LCD monitors use the procedure [[EIZO L660-Host LCD Color Monitor Adjustment](#)].
- For NEC 1850X LCD monitors use the procedure [[NEC 1850X – Host LCD Color Monitor Calibration](#)].
- If the monitor does not have a specific procedure use the procedure for [Generic LCD Monitor Calibration and Gamma set-up](#).

5- SET “GOVERNING BODY”

There are three regulatory bodies that have defined limits for the use of MR systems: The federal Food and Drug Administration (FDA, United States), the International Electrotechnical Commission (IEC, Europe), and the Ministry of Health and Welfare (MHW, Japan). A GEMS MR system that is installed in a country regulated by one of these bodies has to be compliant with the rules they defined. Set the “Governing Body” value for the site as follows:

1. Start the Configuration File Manager on the Service Desktop by selecting **[Utilities]**, **[Config File Manager]**, and then clicking **[Start]**.
2. On the *Config File Manager* screen, select **[MR Config File]**. The *MR Config File* screen is displayed. See Illustration 5-1.



MR CONFIG FILE SCREEN
ILLUSTRATION 5-1

3. Check/change the “Governing Body” selection so it is correct for the country the scanner is located in (**iec**, **mhw1**, **mhw2**, or **fda2**).
4. Click the **[Quit]** button to exit the Config File Manager.
5. Click **[Yes]** in the dialog box question “*Really Quit?*” **If a change was made:** A dialog box (“*MR Config File has been changed.*”) appears. Click the **[Save]** button.
6. Perform a SaveInfo (refer to Section 1-4) to save changes. The install is now done.

6- RECONFIGURING THE SYSTEM USING THE INSTALL GUI & CONFIG FILE MANAGER

To reconfigure pre-loaded software for a new LX install or system upgrade to LX, refer to procedure "Signa Release 8.3 and 8.4 Host Reconfiguration" (OW2SCAE.DOC). If the system needs to be re-configured between software loads, start the Configuration GUI as follows:

1. With Signa up and running, go to the Service Desktop and select **[Install]** on the toolbar. A C-Shell will open and ask for root password ("**operator**" <Enter> is the default). The Install GUI will load and start.
2. Help documentation is built into the GUI. Select **[Help]** on the tab you need information for in the GUI. Select **[Next]** to move through tabs in order. Modify entries to match your system as required.

If you made any changes on the **Hardware** tab, perform the following workaround procedure to avoid losing the site's Isocenter Z calibration value. If you have not made any changes to the **Hardware** tab, or you have not performed Isocenter calibration yet, continue with step 3.

Isocenter Z – Site Value Restoration Procedure:

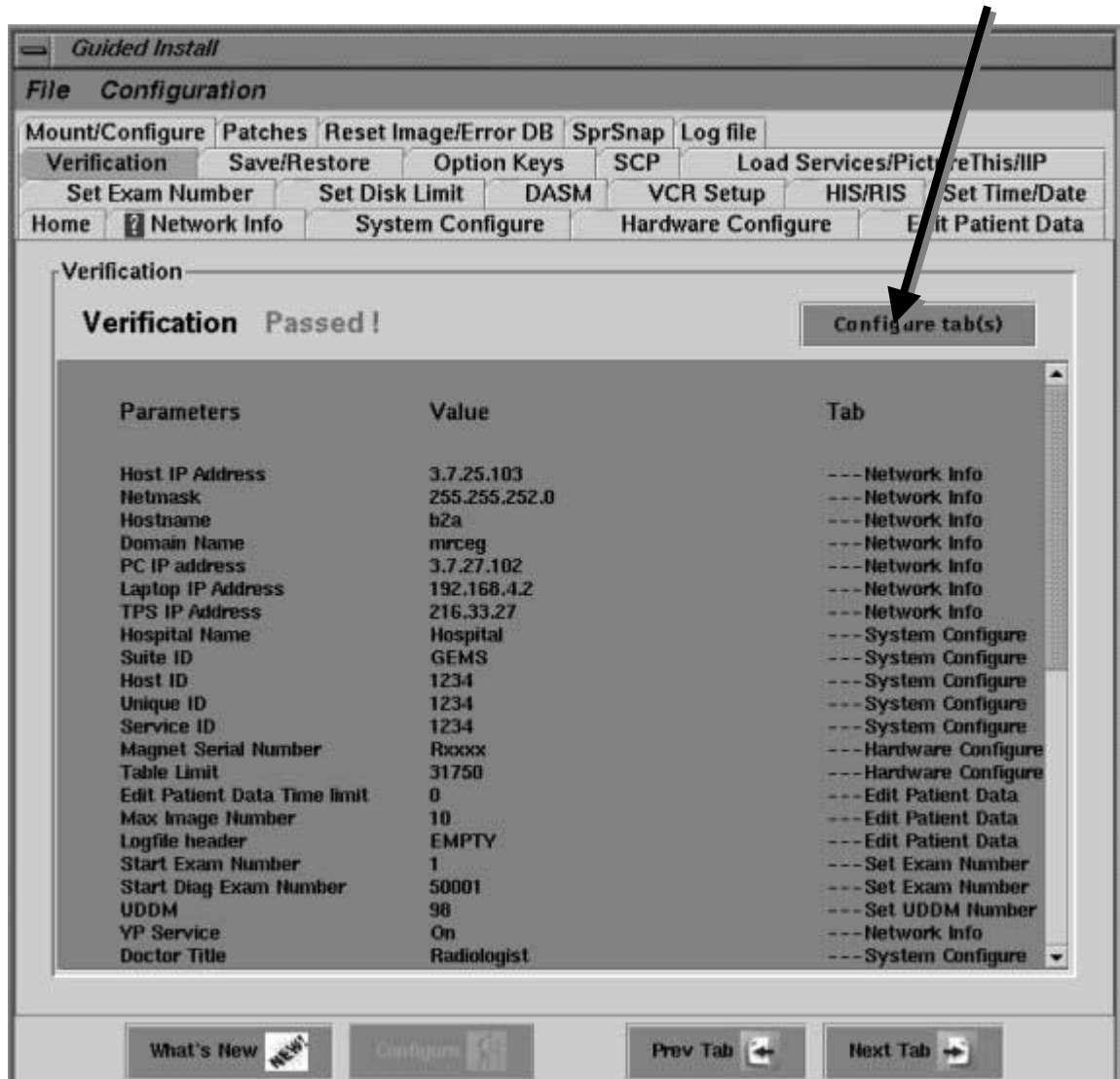
If you made a change on the **Hardware** tab screen. Pressing **[Configure]** or **[Configure tab(s)]** will set the "ISO Vector Z" back to the default value for your magnet type.

To save the calibrated ISO Vector Z value for the site, perform the following:

- a. On the Service Desktop under **[Utilities]**, select **[Config File Manager]**, then **[Start]**. Select the **[MR Config File tab]**, and record the site's "Isocenter Vector Along Z-axis" value. **[Quit]** the Config File Manager tool.
- b. In the Guided Install, press **[Configure]** in the Hardware tab or **[Configure tab(s)]** in the Verification tab.
- c. Re-start the Config File Manager tool, and enter the site Isocenter Vector Z value recorded in step a. **[Quit]** the Config File Manager tool, then select **[Yes]** to save changes.
- d. Continue with Step 4.

(This problem is planned to be fixed on the next software release.)

3. When all changes are made, select **[Configure]** at the bottom of the tab page or wait until all tabs have been modified and go to the **Verification** tab. Verify changes made are correct, then select **[Configure Tab(s)]** to update the system (see Illustration 6-1).



VERIFICATION MENU – CONFIGURE TABS
ILLUSTRATION 6-1

- Before exiting the GUI, insert the Info MOD or new MOD into the drive. Go to the top left corner of the Install GUI and select **<File>** and then **<Save GI Configuration to MOD>**. This process does not create a SaveInfo disk. It just creates a copy of the information already entered in the GUI tabs for use in the next software install.
- Exit the install GUI by selecting **<File>** then **<Quit>**. If any error conditions still exist, you will be warned that no changes will be made before exiting.

Note

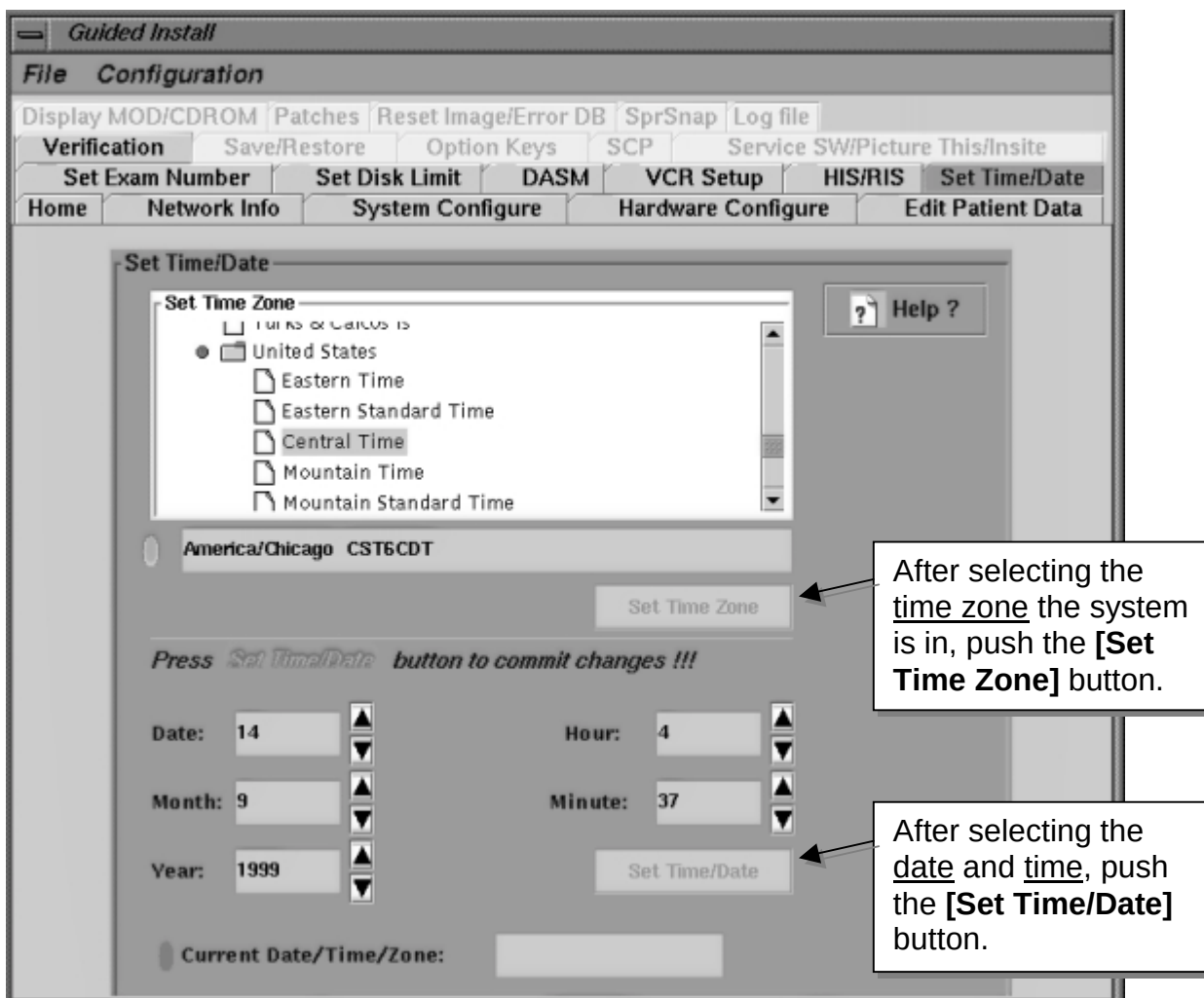
The GUI will not automatically reboot the system in configure mode. It will be up to the user to reboot after making configurations.

- If changes were made to the system configuration, select **[System Shutdown]** and reboot the systems for changes to take effect.
- Perform a SaveInfo on the site's Info MOD to save configuration changes.

7- SET TIME/DATE TAB

The **Time/Date** tab is the only tab that the Configure button will never be highlighted to set options. There are separate buttons for configuring this tab (see Illustration 7-1). If you need to re-configure the Host time and/or date at any time, do the following:

1. Select the Time Zone your system is located in, then select **[Set Time Zone]**.
2. Select the date and local time, then select **[Set Time/Date]**.



SET TIME/DATE MENU
ILLUSTRATION 7-1

APPENDIX A - HOST CONFIGURATION ENTRY DESCRIPTIONS

Host IP Address: The Host IP address should be obtained from the site network administrator if the scanner will be connected to the sites internal network. Or obtain an address by calling the OnLine Center (OLC) MR Support.

IWS PC IP Address: The IWS PC IP address should be obtained from the site network administrator if the scanner will be connected to the sites internal network. Or obtain an address by calling the OnLine Center (OLC) MR Support.

FE Laptop IP Address: The Laptop IP address should be obtained from the site network administrator if the scanner will be connected to the sites internal network. Or obtain an address by calling the OnLine Center (OLC) MR Support.

TPS Subnet Address: The TPS is on a subnet of the Signa LX scanner. A default address is loaded during the load from cold process and it should not be changed.

Hostname: This is the network name given to the scanner host computer.

Hospital Name: The Name of the hospital or other facility where the Signa LX is installed. Name can include characters and/or integers up to 32 characters.

Doctors Title: The title to be used at the scanning site by the reading physician (Radiologist or Diagnostician).

Suite ID: Name of the Suite (if any) the scanner will be connected to (name should be at least four characters long). Leave blank otherwise.

Unique ID: The Unique System ID is a 16 alphanumeric character string assigned at the time the system is first installed. The Unique system ID is assigned by GE CARES when the first dispatch is opened to document the start of installation.

Service ID: The Service ID is a 16 alphanumeric character string assigned at the time the system is first installed. This is the same number that GE CARES uses to track financial and administrative data for the system.

Weight Unit: The weight system to be used by the scanner. Kilo Grams or US pounds.

Monitor type: The type of monitor the scanner will be using.

Windows 95 OEM Number: The Microsoft OEM Number found on the "Proof of Authenticity" for Windows 95. _____

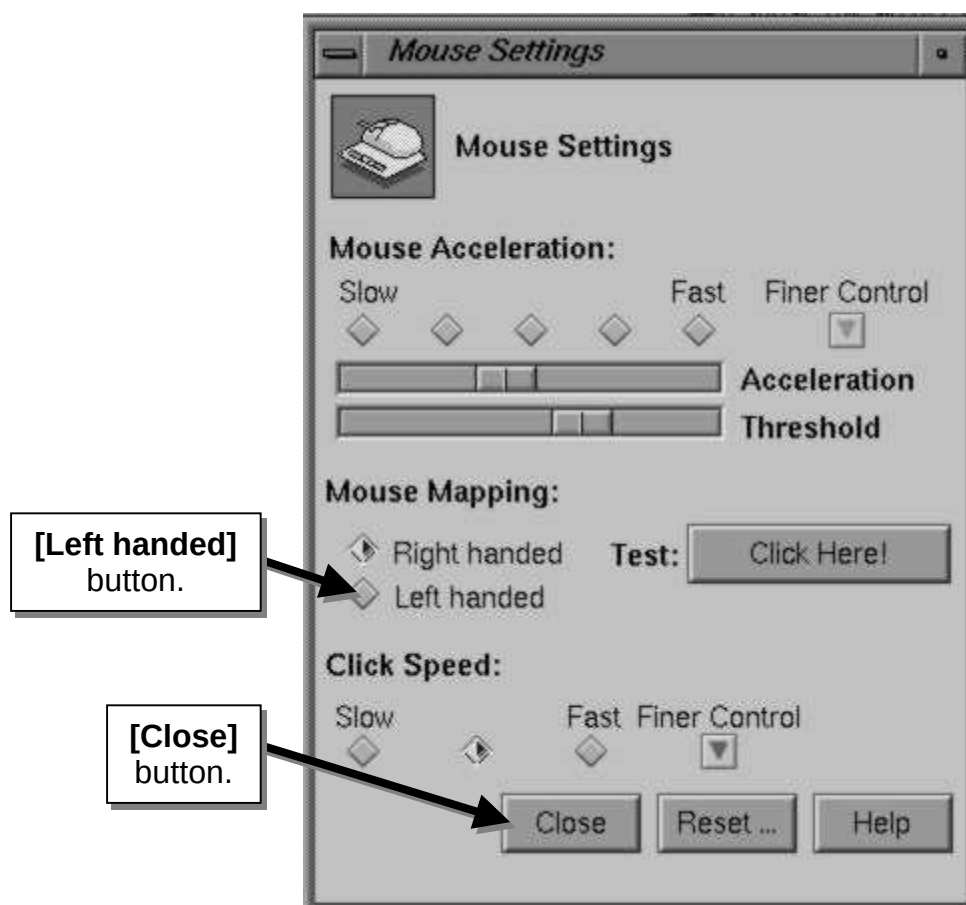
NetMask: For new installs, the Netmask should be obtained from the site network administrator if the scanner will be connected to the site's internal network. If this is a stand-alone system, use the default netmask.

APPENDIX B - CONFIGURING MOUSE FOR LEFT HANDED USE

The system Host mouse is default configured for right handed use. It can be configured for permanent or temporary left handed use.

B-1 Configuring Mouse For Permanent Left Handed Use

1. If Signa is up and running, you must first log out and bring the system up to the Login screen. Otherwise, turn the system on and proceed to the Login screen.
2. Log in as **root**. The system default password is "**operator**". On the root desktop, use the mouse cursor on the ToolChest and click and slide through **[Desk Top]** then **[Customize]** then **[Mouse]**.
3. Select **[Left Handed]** under Mouse Mapping, then select **[Close]** to accept the changes. See Illustration B-1.



SET TIME/DATE MENU
ILLUSTRATION B-1

4. Log out of the root desktop and reboot the system. To change the mouse mapping back to Right handed. Repeat this section but select **[Right Handed]**, then reboot the system.

B-2 Configuring Mouse For Temporary Left Handed Use

Using this method will change the mouse mapping from right handed to left handed, but only until Signa is rebooted, when it will revert back to right handed mapping.

1. If not already at the Signa scanning level, go there. Open a command window and on the command line type: **mouse <Enter>**.
2. Select **[Left Handed]** under Mouse Mapping, then select **[Close]** to accept the changes. (See Illustration B-1.)
3. To reconfigure the mouse back to right handed operation, repeat this section but select **[Right Handed]**, or reboot Signa.

APPENDIX C - DISK DRIVE LOW LEVEL FORMAT

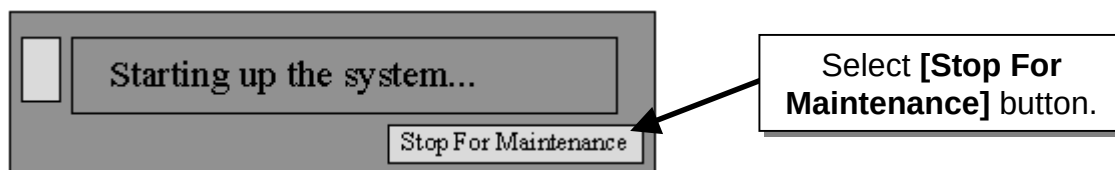
This section describes how to do a low-level format of the disk drives. A low-level format should only be done if loading a different Operating System (OS) or if suspected bad blocks on a disk drive are causing system problems. Depending on the system and disk drives, this procedure can last up to 1 hour per drive (Indigo) or as little as 20 minutes per drive (Octane).

1. Insert the **IRIX 6.5** CD labeled "**Boot/Update**" into the CDROM Drive on the workstation (right side).

Note

User responses are bolded text

2. With the system up and running, go to the service desktop and select [**Shutdown**]. If the system is not up and running, switch on power at this time.
3. When prompted, click on [**Restart**]. Stop the reboot process by clicking on [**Stop For Maintenance**] when it appears.



4. Next, click on [**Enter Command Monitor**] when it becomes available. Type one of the following depending on the system your working on:

For SGI Octane computer type:

```
>> boot -f dksc(1,3,8)sash64 dksc(1,3,7)stand/fx.64 --x <Enter>
```

For SGI Indigo computer, type:

```
>> boot -f dksc(0,3,8)sashARCS dksc(0,3,7)stand/fx.ARCS --x <Enter>
```

5. The rest of the install does not depend on the type of SGI computer (Indigo/Octane). Continue typing as follows:

```
fx: "device-name" = (dksc) <Enter>
```

```
.
```

```
fx: ctrlr# = (0) <Enter>
```

```
.
```

```
fx: drive# = (1) <Enter> or 2 <Enter> if doing the second drive.
```

```
.
```

```
fx: d/fo <Enter>
```

```
.
```

```
Press <Enter> to accept current parameters.
```

```
.
```

```
Type: Yes <Enter> to start the format.
```


6. The format can take up to 20 minutes per disk drive for an Octane computer. There will be no prompts to the user while the format is running. Be patient and do not interfere with the system. When the format is finished the system will prompt with:

Format completed successfully

.

.

fx/debug>

7. Reset or power cycle the computer to exit debug and proceed to Section 2 of this document to proceed with the OS (Operating System) load for the system.

APPENDIX D - GENERAL INFORMATION AND NICE TO KNOWS

D-1 Bore Temperature Sensor - Must Be Working!

If the bore temperature sensor is disabled, sites with iDrive software will have problems. Sites running the iDrive software must have a functional bore temperature sense circuit. The symptom for a non-functional Bore sensor will be evident when an iDrive scan is attempted; the prescan will complete OK, but the scan will stop immediately after the first pulse is heard in the body coil. No images will be produced and no errors will appear in the log.

If the system has iDrive software (as listed in *Option Key* tab of Guided Install), to verify functional bore temperature circuit, set up the scan in Table D-1 (or any other Real Time scan). Download the scan and select **[Scan]**. If prescan completes and the scan continues to run, you have a functional bore temperature circuit. If prescan completes OK, but the scan stops immediately after the first pulse, check the bore temperature circuit.

TABLE D-1
IDRIVE SAMPLE PROTOCOL

<u>PATIENT REGISTER</u>	[New Pt]	<u>SCANNING RANGE</u>	
<u>PATIENT INFORMATION</u>		FOV.	[32]
Patient Id	geservice	Slice Thickness.	[7]
Patient Name	test	Spacing	[1.5]
Weight (Lb)	111	Start	0
	[Landmark]	End..	0
Landmark	[Sternal Notch]	# Slices	1 (default)
		L/R Center	0 (default)
		P/A Center	0 (default)
<u>PATIENT PROTOCOLS</u>	[Patient Position]	Table Delta	0.00 (default)
<u>PATIENT POSITION</u>		<u>ACQUISITION TIMING</u>	
Patient Position	[>] [Supine]	Freq	[128]
Patient Entry.	[>] [Head First]	Phase	[128]
Coil	[...] [Body] [Accept]	NEX	[1]
		Phase FOV	1.00 (default)
<u>IMAGING PARAMETERS</u>		Freq Dir	[>] [R/L]
Plane	[>] [Axial]	Auto Center Freq	[>] [water]
Mode	[>] [2D]	Autoshim	[on]
Pulse Seq	[...] [FGRE] [Accept]	(lowest window)	[Save Series]
Imaging Options	[Real-time]		
Psd Name	no entry		
Protocol	no entry		
<u>SCAN TIMING</u>			
# of Echoes	1 (default)		
TE	[Mn Full]		
TR.	[Mn]		
Flip Angle	[20]		
Bandwidth	[62.50]		

D-2 Signa Power up Delay

When Signa is first powered on there is a considerable period of time that nothing is seen on the monitor; up to 4 minutes depending on the computer type. The system is checking configurations and starting software processes at this time. If you are concerned that the system is hung, push the <Esc> button on the top left side of the keyboard. The system will then post to the monitor its entire boot up activities.

D-3 "Configup" Script Modifications To Files

The "configup" script updates the MR System configuration file with new changes as needed:

1. The "configup" script will add file revision identifiers to the MRconfig.cfg and GradientConfig.cfg files. The revision is added as "fileFormatRev = "xx.xx" (e.g. 84.M3).
2. The "configup" script updates the MRconfig.cfg file with the new "fastprescan" parameter if it is not present (if "fastprescan" parameter is present, it does not add or change it):

```
# This field is added by shivar@wge for MRIge47785
# A fastprescan field to turn Faster Prescan on & off .
#           type = integer, legal values = 0 (off)
#                                           = 1 (on)
fastprescan = "1"
```

3. The "configup" script updates the MRconfig.cfg file with the new "cerdType" parameter if it is not present (if "cerdType" parameter is present, it does not add or change it):

```
# Type of (Universal) Combined Exciter Receiver
# Data Acquisition Board (CERD)
# type = integer
# Usage
# X Y Y
# \ \ \----- (U) CERD Slot #1 (VME slot 13 , right next to
# power supply)
# \ \----- (U) CERD Slot #2 (VME slot 10)
# \----- MKE or YMS Receiver mapping
# Where X =
# 0: MKE receiver mapping scheme
# 1: YMS receiver mapping scheme
#
# Where Y =
#
# 0 : No(U) CERD Installed
# 1 : Narrowband CERD
# 2 : Universal CERD (UCERD)
# Legal Combinations
#
# 0yy: MKE default receiver mapping
# 1yy: YMS default receiver mapping
# x01: Narrowband CERD on Slot #1
# x02: Universal CERD on Slot #1
# x21: Narrowband CERD on Slot #1 ;Universal CERD on Slot #2
```

```
# x22: Universal CERD in both Slot #1 and Slot #2
cerdType = "001"
```

4. The "configup" script updates the MRconfig.cfg file with the new maximum Bandwidth parameters for systems with multiple Receivers if not present (if the following parameters are present, it does not add or change them):

```
# Added by chakraba@mr.med.ge.com on Aug. 23, 1999 SPR #MRIge54996
maxBW2 = "125.0"
maxBW3 = "125.0"
maxBW4 = "125.0"
```

5. The "configup" script updates the GradientConfig.cfg file with the new Full Power SGD parameters if not present (if the following parameters are present, it does not add or change them):

```
# Added by shivar@wge for MRIge49945
Sdlim = "0.60"
Lcoil = "1.40"
Rac = "152"
Pmgs = "3000"
mGRAMpower = "9000"
```

6. **CRM sites only:** The "configup" script updates the GradientConfig.cfg file with three spiral PSD gradient group delays (if the following parameters are present, it does not add or change them):

```
#-----#
# Added by araballi@mr for CAI #
#-----#
psdGradDelayX = "180.000000"
psdGradDelayY = "180.000000"
psdGradDelayZ = "180.000000"
```

D-4 "Coilup" Script Modifications To Coil Configuration File

When RestoreINFO is run after loading software the first time, the "allup" script will invoke the "coilup" script to update all coil configuration files as follows.

1. The coilup script begins with a copy of the default 8.4 CoilConfig.cfg file that matches the site's system configuration. This contains **all** GE-supported coil definitions.
2. Merge customer coils (non-GE) in the CoilConfig.cfg file from the InfoMOD MOD into the default 8.4 CoilConfig.cfg file.
3. Update the "reconScale" parameter for head and body coils to match the value in the CoilConfig.cfg file from InfoMOD.
4. Check version number of the CoilConfig.cfg file from the InfoMOD to determine if upgrade is necessary.

5. If upgrade needed, modify files by adding a new parameter “headDefaultFreqDir” to all the customer coils. For all coils, including customer coils, headDefaultFreqDir should be set to 0 except as listed in Table D-2 below; for those coils in the table headDefaultFreqDir should be set to 1. This new parameter defines the default Frequency Direction for the coil. 1 = same default frequency direction as a head coil. Table D-3 lists frequency direction defaults for coils.

TABLE D-2
COILS WITH DEFAULT FREQ DIRECTION SAME AS HEAD COIL

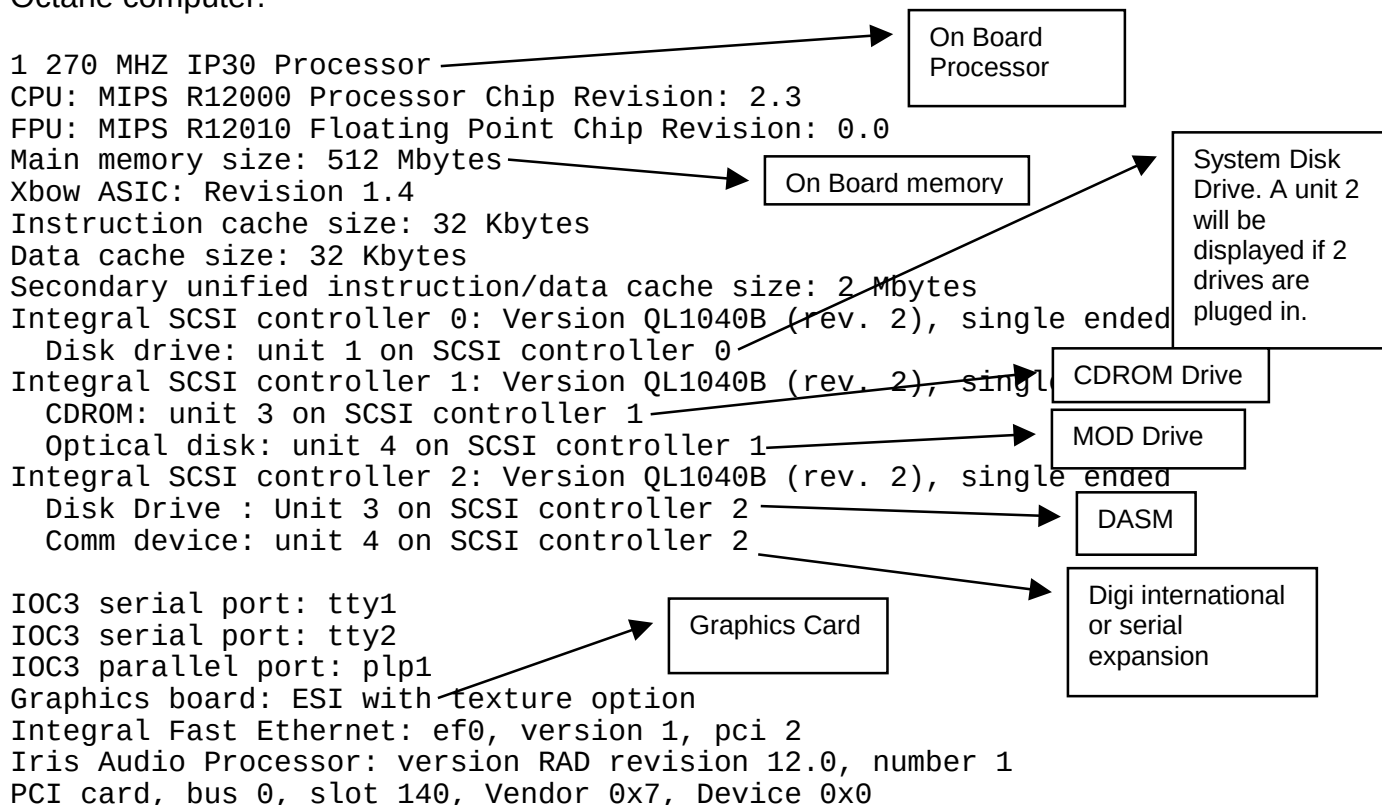
<u>T/R</u>	<u>Receive Only</u>
Head (GE)	Quad Extremity (GE)
Linear Knee (GE)	Receive Only Knee (GE)
Quad Knee, 1.0T (GE)	Receive Only NV Array (GE)
NV Coil (Medrad)	Receive Only NV Coil (Medical Advances)
T/R Knee Array (Medrad)	

TABLE D-3
DEFAULT FREQUENCY DIRECTION FOR COILS

	<u>“Head-type” coil</u>		<u>Other (Body, Surface) coil</u>	
	<u>Standard</u>	<u>EPI</u>	<u>Standard</u>	<u>EPI</u>
Ax	A/P	R/L (safety)	R/L	R/L
Sag	S/I	S/I	S/I	S/I
Cor	S/I	R/L (IQ [ghosting])	S/I	S/I
Obl.	Must follow table above for “most-like” plane			

D- 5 [hinv] Command

The **[hinv]** command can be used to check the configuration of the operator workspace. To use the command, open a command window and at the prompt type: **hinv <Enter>**. The following is an example of an **[hinv]** output of a common configured Signa system using Octane computer.



APPENDIX E - CONFIGURING A MOBILE SYSTEM

Description

The Mobile Site Setup feature provides the ability to change a mobile's Hospital Name, Hostname, IP Address, Netmask, and the Host or Network DICOM information at the Mobile Site Setup screen. Mobile site information is saved with **Save Info**.

Background

To access the Mobile Site Setup, you must first logout of Signa or perform a Signa Shutdown. At the Login screen, type **mobile** for the user name.

The screenshot shows the 'Mobile Site Setup' window. At the top, the title bar says 'Mobile Site Setup'. The main title 'Mobile Site Setup' is in large blue font. Below it, the section 'MR Scanner' is highlighted. Callout 1 points to the 'Hospital Name' text box. Callout 2 points to the 'Hostname' text box. Callout 3 points to the 'IP Address' text box. Callout 4 points to the 'Netmask' text box. Callout 5 points to a list box containing 'St Lukes', 'Columbia Memorial', 'United', and 'Mt Sinai'. Below this, callout 6 points to the 'Host DICOM Information' section. Callout 7 points to the 'Network DICOM Devices' section. Callout 8 points to the 'Add' button. Callout 9 points to the 'Change' button. Callout 10 points to the 'Remove' button. Callout 11 points to the 'Search' button. Callout 12 points to the 'Activate' button. Callout 13 points to the 'Done' button. The 'Host DICOM Information' section has 'AE Title' and 'Port No' text boxes. The 'Network DICOM Devices' section has three columns: 'IP Address', 'AE Title', and 'Port No', each with five empty text boxes.

MOBILE SITE SETUP SCREEN
ILLUSTRATION E-1

Mobile Site Setup Screen Description (See Illustration E-1)

Hospital Name

Enter the hospital name (up to 30 characters using A-Z, a-z, 0-9, & - only).

Hostname

The system's host name can consist of up to 30 characters, and is stored in the configuration file. A GE Service Engineer enters the system's host name initially.

IP Address

An IP Address is four integers with three dots between them that uniquely identifies the computer on the network. Consult a GE Service Engineer for this information.

Netmask

Netmask should be in hex and in the following format: 0xhhhhhhhh, where h is the hex digit. Consult a GE Service Engineer for this information.

List of Previously Entered Hospitals

Select this key to choose a different hospital.

AE Title and Port No.

Host DICOM information is optional for mobiles that interface with a Network DICOM device. Consult a GE Service Engineer for this information.

Network DICOM Devices

Used to add up to 5 Network DICOM Devices that interface with the mobile. Consult a GE Service Engineer for a list of IP Addresses, AE Title and Port No.

Add

Select **Add** to add a hospital to the list.

Change

Select **Change** to edit or modify previously stored hospital information.

Remove

Select **Remove** to delete a hospital from the list.

Search

Select **Search** to find a hospital on the list. Search options are executed by either hospital or hostname.

Activate

Select **Activate** to activate the selected hospital.

Done

Select **Done** to exit the Mobile Site Setup screen.



MOBILE SITE ACTIVATION SCREEN
ILLUSTRATION E-2

Mobile Site Activation Screen Description (See Illustration E-2)

Activate Site

Select this button to choose a different hospital. A reboot is required with this selection.

Get Active Site

Select this button to use the currently active hospital and its network and DICOM information. A reboot is not required.

Cancel

Select this button to cancel a selection and close the Mobile Site Activation pop-up window.

Select **Get Active Site** to obtain the currently active hospital's configuration hostname, IP Address, Netmask and the Host or Network DICOM information for the Mobile Site Setup screen.

Applications

This feature is used to update the image annotation with the appropriate hospital or outpatient facility's name.

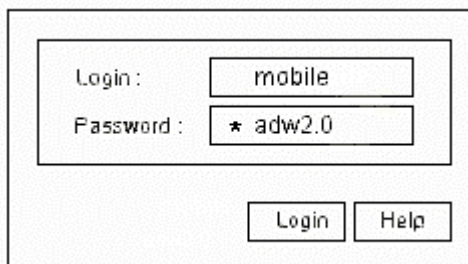
Considerations

- Purchased options and features stay with the mobile unit and will not change from site to site.
- To add a hospital to the list, the minimum entry is the MR Scanner information (Hospital, Hostname, IP Address and Netmask).
- Consult with a GE Service Engineer before entering the hostname, IP Address, Netmask, Host, or Network DICOM information.
- Periods, commas, or apostrophes (non-alphanumerics) are not valid for hospital names. For example: St. Luke's is invalid whereas St Lukes is valid. To add a new hospital, select **Get Active Site**, and the Mobile Site Setup screen prompts for information to be entered into the mobile database. A response of Yes will add the hospital. If the Mobile Site Setup screen does not provide a prompt, the hospital name is already in the mobile database.
- If changes are made to DICOM or Network information in the Browser or Install, the current hospital and Get Active Site need to be deleted.
- Only **Activate Site** requires a reboot of the system.

Procedure

To Reduce Setup Time (by eliminating the reboot) at the Next Hospital:

1. Perform a system logout.
2. Select **mobile** and press Enter at the Login screen.
3. Type in **adw2.0** and press Enter at the Login screen.

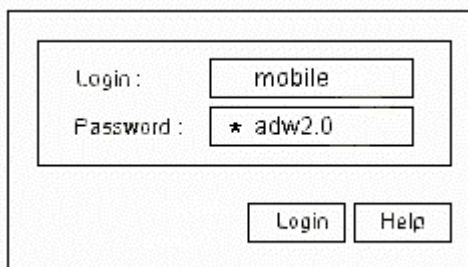


A screenshot of a login interface. It has two input fields: 'Login:' with 'mobile' entered, and 'Password:' with '★ adw2.0' entered. Below the fields are two buttons: 'Login' and 'Help'.

4. Select the new hospital.
5. Select **Activate, Done** and power down.
6. At the next power-up, the system will display the selected hospital name.

To Select a Different Hospital

1. Select **mobile** and press Enter at the Login screen.
2. Type in **adw2.0** and press Enter at the Login screen.



A screenshot of a login interface, identical to the one above. It has two input fields: 'Login:' with 'mobile' entered, and 'Password:' with '★ adw2.0' entered. Below the fields are two buttons: 'Login' and 'Help'.

*(Actual password does not appear on the screen).

3. Select a hospital from the list at the Mobile Site Setup screen.

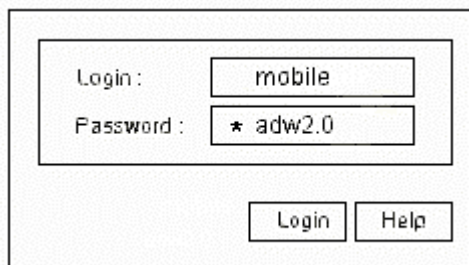


A screenshot of a screen displaying a list of hospital names: 'St Lukes', 'Columbia Memorial', 'United', and 'Mt Sinai'. There is a vertical scrollbar on the right side of the list.

4. Select **Activate** at the Mobile Site Setup screen.
5. Select **Activate Site** at the Mobile Site Activation pop-up window. The mobile GUI will now display the selected hospital's information.
6. Select **reboot** and the system will reboot and activate the selected site.
7. Select **signa** and press Enter at the Login screen.
8. Type in **adw2.0** and press Enter at the Login screen.

To Add a New Hospital Using Get Active Site:

1. Select **mobile** and press Enter at the at the Login screen.
2. Type in **adw2.0** and press Enter at the Login screen.



The screenshot shows a login interface with two input fields. The first field is labeled 'Login :' and contains the text 'mobile'. The second field is labeled 'Password :' and contains the text '★ adw2.0', where the asterisk indicates that the password is masked. Below the input fields are two buttons: 'Login' and 'Help'.

*(Actual password does not appear on the screen).

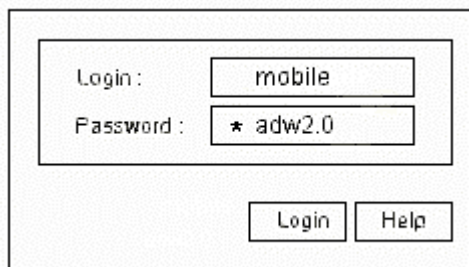
3. Select **Activate** at the Mobile Site Setup screen.
4. Select **Get Active Site** at the Mobile Site Activation pop-up menu. The Mobile Site Setup screen will update using the currently active hospital's information.

If the hospital name is not in the mobile database, the system will ask if you want to add the hospital information. Select **Yes**, to add the hospital into the mobile database.

- a. Select **Activate** at the Mobile Site Setup screen.
- b. Select **Activate Site** at the Mobile Site Activation pop-up menu. The mobile GUI will now display the selected hospital's information.
- c. Select **reboot** and the system will reboot and activate the selected site.
- d. Select **signa** and press Enter at the Login screen.
- e. Type in **adw2.0** and press Enter at the Login screen.

To Add a New Hospital

1. Select **mobile** and press Enter at the Login screen.
2. Type in **adw2.0** and press Enter at the Login screen.



This screenshot is identical to the one above, showing the login screen with 'mobile' in the Login field and '★ adw2.0' in the Password field, with 'Login' and 'Help' buttons at the bottom.

*(Actual password does not appear on the screen).

3. Select **Add** at the Mobile Site Setup screen.
4. Type in the Hospital name, Hostname, IP Address and Netmask at a minimum. Consult with a GE Service Engineer for this information if unknown.
5. Select **Activate** at the Mobile Site Setup screen.
6. Select **Activate Site** at the Mobile Site Activation pop-up menu. The mobile GUI will now display the selected hospital's information.
7. Select **reboot** and press Enter at the Login screen.
8. Select **signa** and press Enter at the Login screen.
9. Type in **adw2.0** and press Enter at the Login screen.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	Nov 24, 1999	M. Keber	Initial version for Release 8.4 ME1 software.
B	Jan 17, 2000	M. Keber	Added new Cardiac Option keys, updated Appendix D for CRM configup changes, and made minor changes based on ME2 validation.
0	May 10, 2000	M. Keber	Updated per SPRs MRIge58871, MRIge58766, MRIge59627, and MRIge59694, restructured Section 3 into subsections for MR Apps, Video Driver, Service Software, Options, added Appendix D-4 and Appendix E, incorporated bay validation markups and minor 8.4 M3 changes, removed old 8.2.5 and 8.3 information.
1	Sept 28, 2000	M. Keber	Updated for minor M4 changes and per SPR MRIge60849 to add "Governing Body" check for new installations. Added reference to Host Reconfiguration procedure for new installations and upgrades to LX.
2	April 26, 2001	Hawthorne	Updated section 4 to add gamma settings for Eizo LCD monitor.
3	Oct 26, 2001	Hawthorne	Updated section 4 gamma settings to point to needed document to adjust gamma. Added hinv information is appendix D updated information for IVI protocols in section 3
4	Nov 13, 2001	Hawthorne	Updated command prompt for loading Video drivers in section 3-2 step 4 and added a note suggesting a [ProDiag] configuration in section 3-3-3.
5	Nov 13, 2001	Hawthorne	Moved IVI, Functool, Interactive Vascular Imaging and Clairview option load earlier in the procedure to section 3-1 step 10 to prevent customer protocol load problems. Added InSite Checkout Model type to section 3-3-3.
6	July 11, 2003	D. Hofstetter	Fixed the Title of section 5 to solve a cqa